

6.2 2p2z controller

The bilinear transform is applied to the s-domain Type II compensator by replacing every instance of 's' in the s-domain transfer function with the bilinear mapping. The initial substitution is given in 18 and the simplified result is given in 19.

Type II compensator initial z mapping

$$H_c[z] = \left(\frac{\omega_{CP0}}{\frac{2}{T_s} \frac{1-z^{-1}}{1+z^{-1}}} \right) \frac{\left(\frac{\frac{2}{T_s} \frac{1-z^{-1}}{1+z^{-1}}}{\omega_{CZ1}} + 1 \right)}{\left(\frac{\frac{2}{T_s} \frac{1-z^{-1}}{1+z^{-1}}}{\omega_{CP1}} + 1 \right)} \quad (18)$$

Type II compensator initial z mapping

$$H_c[z] = \frac{B_2 z^{-2} + B_1 z^{-1} + B_0}{-A_2 z^{-2} - A_1 z^{-1} + 1} \quad (19)$$

In the discrete-time z-domain, the transfer function now has two z-domain poles and two z-domain zeros. Therefore, this controller is now referred to as a two-pole two-zero controller or 2p2z for short.

The numerator of the transfer function has been grouped into like terms consisting of 'B' coefficients and likewise, the denominator has been grouped into like terms consisting of 'A' coefficients. These coefficients are given in 20 to 24.

Type II compensator B0 term

$$B_0 = \frac{T_s \times \omega_{CP0} \times \omega_{CP1} \times (2 + T_s \times \omega_{CZ1})}{2 \times (2 + T_s \times \omega_{CP1}) \times \omega_{CZ1}} \quad (20)$$

Type II compensator B1 term

$$B_1 = \frac{T_s^2 \times \omega_{CP0} \times \omega_{CP1}}{2 + T_s \times \omega_{CP1}} \quad (21)$$

Type II compensator B2 term

$$B_2 = \frac{T_s \times \omega_{CP0} \times \omega_{CP1} \times (-2 + T_s \times \omega_{CZ1})}{2 \times (2 + T_s \times \omega_{CP1}) \times \omega_{CZ1}} \quad (22)$$

Type II compensator A1 term

$$A_1 = \frac{4}{(2 + T_s \times \omega_{CP1})} \quad (23)$$

Type II compensator A2 term

$$A_2 = \frac{-(2 - T_s \times \omega_{CP1})}{(2 + T_s \times \omega_{CP1})} \quad (24)$$

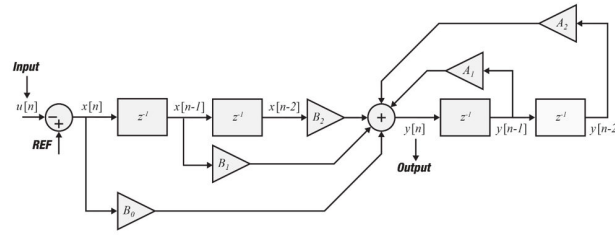
There is no need to calculate these controller coefficients by hand as the software tool Biricha ST PLD available from www.biricha.com performs these calculations for the user.

Finally, now that the controller is in the discrete-time 2p2z form, it can be converted into a linear differential equation, which can be easily implemented on the device. This takes advantage of the shifting property of the z-domain transfer function and converts from z-domain to discrete sample in 25.

Type II compensator linear differential equation

$$y[n] = A_1 y[n-1] + A_2 y[n-2] + B_0 x[n] + B_1 x[n-1] + B_2 x[n-2] \quad (25)$$

Where $y[n]$ is the output for the current sampling interval. The structure of this controller is depicted in Figure 19.

Figure 19. 2p2z controller structure


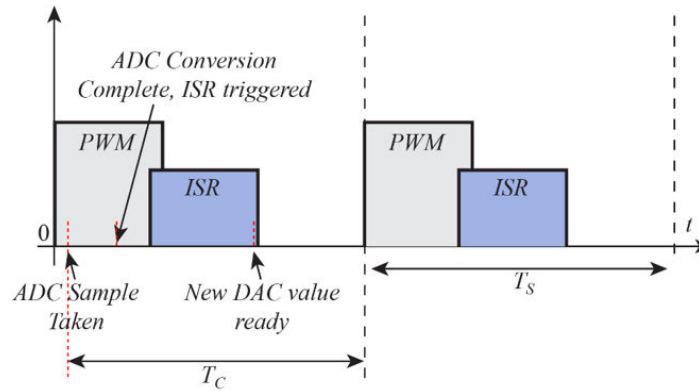
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For the fast current loop of this average current mode converter, the output of the controller during this sampling interval would be the new demand average current value that is used in the following switching period. Therefore, there is now an equation, which can implement the analytically designed controller and is calculated by the device at every sampling interval when there is a new sample available.

6.3 Pure time delays

In this discrete-time digital system, there are additional time delays, which are not present in the equivalent analog continuous-time system. A pure time delay in the discrete-time system results in a phase delay that is proportional to the frequency and time delay.

There are two sources of time delay in the digital system. The first is the time from which the output voltage sample was taken, to the time at which it was used. This is referred to as the calculation delay. In effect, the older the sample the more phase delay is associated with the sample. For example, a convenient time to trigger the ADC and sample the output voltage and inductor current is in the middle of the PWM switching cycle. This ensures that the average current is being sampled away from the switching edges. In this case, the voltage is sampled, converted, and used in the controller to provide the new value of duty. This sampling and calculation time may take several hundred nanoseconds and the new demand peak current value is ready to use towards the end of that switching period. This situation is described in Figure 20.

Figure 20. Sources of delay within the discrete-time system


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However, as the new duty cycle value is not used until the following switching cycle, the total time delay for this calculation time is considered to be half of the switching period.

The second contribution to the time delay is that of the zero-order-hold introduced by this sampled data system. The ADC is triggered once per switching cycle, this sampled data is used to calculate the new value of duty cycle value and that duty cycle value is then held constant for the remainder of the next switching period by the HRTIM module. Therefore, there is an additional zero-order-hold in this system. The frequency response of a ZOH is given in 26.

Zero-order-hold transfer function

$$F_{ZOH}(j\omega) = \frac{1 - e^{-j\omega T_s}}{j\omega} \quad (26)$$

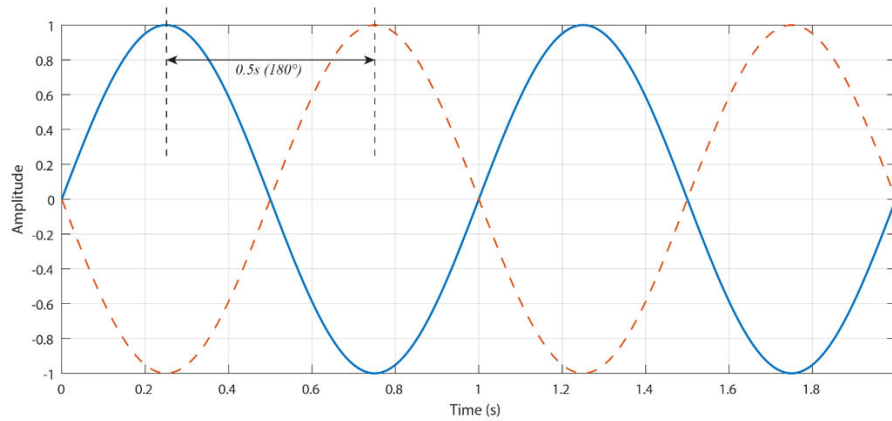
This can be shown to be a complex number expressed in polar form, and therefore the phase angle is given in 27.

Zero-order-hold phase angle

$$\angle F_{ZOH}(j\omega) = -\omega \frac{T_s}{2} \quad (27)$$

To understand how this phase angle can relate to a time delay, consider as an example, the 1 Hz sine wave shown in Figure 21. If this sine wave were delayed by 0.5 seconds, the result would be the sine wave shown as a dashed line in Figure 21.

Figure 21. Time delay relationship to phase delay



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It is clear from this that the time-delayed sine wave now has a phase delay of 180°. Therefore, the equation to calculate the phase delay of a sine wave given the pure time delay T_D is given in 28.

Phase delay relationship

$$p.d. = -2\pi f T_D \quad (28)$$

This indicates that there are some phase delay across all frequencies, however, when $T_D < < \frac{1}{f}$ the phase delay is negligible. This is true for very low frequencies assuming that the phase delays in the digital system can be kept to a minimum. The frequency at which this phase delay could become a concern would be the crossover frequency of the open-loop system, as this is where the phase margin is defined. Therefore, the amount of phase loss at the crossover frequency can be calculated, given the known time delays in the digital system.

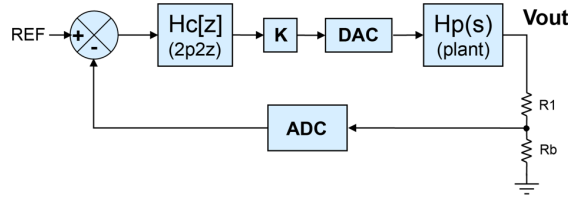
Phase erosion at crossover frequency

$$p.e. = -2\pi f_x T_D \quad (29)$$

The expected phase erosion at the crossover frequency can be added to the desired phase margin such that this phase loss is compensated for at design time. This can be achieved by adding the expected phase erosion to the phase margin specified in 29. This adjusts the location of the compensator zero to add the require phase boost and compensate for this phase loss such that the resulting overall phase margin meets the specification.

6.4 Digital gains

The compensator poles and zeros have been analytically calculated to achieve the specified crossover frequency and phase margin. If the additional gains within the digital system are not accounted for, then the overall loop gain, and thus phase margin, is incorrect.

Figure 22. Additional gains within the digital system


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Consider the block diagram of the digital control loop shown in Figure 22. The output voltage, V_{out} , need to be scaled down to a voltage suitable for sampling by the ADC onboard the microcontroller. This is the purpose of the potential divider formed by R_1 and R_b . This also determines the setpoint value that the controller must regulate to. The gain of the pre-ADC scaling potential divider can be calculated in 30.

Pre-ADC gain

$$G_{PD} = \frac{R_b}{R_1 + R_b} \quad (30)$$

Then, the ADC onboard the microcontroller converts the voltage, which is between 0 V and 3.3 V, and provide a 12-bit output, a number between 0 and 4095. Therefore, the gain of the ADC can be calculated using 31.

ADC gain

$$G_{ADC} = \frac{2^{12} - 1}{3.3V} \quad (31)$$

The output of the controller is used to update the DAC register, which sets the demand peak current level for the internal comparator. The 12-bit DAC gives an output voltage of 3.3 V for an input of 4095. Therefore, the DAC gain can be calculated using 32.

DAC gain

$$G_{DAC} = \frac{3.3V}{2^{12} - 1} \quad (32)$$

To achieve the correct loop gain, the controller must negate these additional gains. That is achieved using the scaling factor K , which can be calculated in 33.

K scaling factor

$$K = \frac{1}{G_{PD}G_{ADC}G_{DAC}} \quad (33)$$

Finally, the ADC output is compared with the digital reference set point. This digital reference must be in the same scaling as the ADC output and consider the divider on the input of the ADC. Therefore, the reference can be calculated using 34.

REF scaling factor

$$REF = V_{OUT}G_{PD}G_{ADC} \quad (34)$$